```
import pandas as pd
```

```
df=pd.read_csv('/content/covid_toy.csv')
```

```
df.head()
```

	age	gender	fever	cough	city	has_covid
0	60	Male	103.0	Mild	Kolkata	No
1	27	Male	100.0	Mild	Delhi	Yes
2	42	Male	101.0	Mild	Delhi	No
3	31	Female	98.0	Mild	Kolkata	No
4	65	Female	101.0	Mild	Mumbai	No

Next steps: Generate code with df | New interactive sheet

```
df.isnull().sum()
```

	0
age	0
gender	0
fever	10
cough	0
city	0
has_covid	0

dtype: int64

```
df['fever']=df['fever'].fillna(df['fever'].mean())
```

```
df.isnull().sum()
```

	0
age	0
gender	0
fever	0
cough	0
city	0
has_covid	0

dtype: int64

```
df['fever'].mean()
```

```
np.float64(100.84444444444445)
```

```
df.head(2)
```

	age	gender	fever	cough	city	has_covid
0	60	Male	103.0	Mild	Kolkata	No
1	27	Male	100.0	Mild	Delhi	Yes

Next steps: Generate code with df | New interactive sheet

```
df['gender'].value_counts()
```

gender	count
Female	59
Male	41

dtype: int64

```
df['gender']=df['gender'].map({'Male':0,'Female':1})
```

```
df.head(7)
```

	age	gender	fever	cough	city	has_covid
0	60	0	103.000000	Mild	Kolkata	No
1	27	0	100.000000	Mild	Delhi	Yes
2	42	0	101.000000	Mild	Delhi	No
3	31	1	98.000000	Mild	Kolkata	No
4	65	1	101.000000	Mild	Mumbai	No
5	84	1	100.844444	Mild	Bangalore	Yes
6	14	0	101.000000	Strong	Bangalore	No

Next steps: Generate code with df | New interactive sheet

```
df['cough'].value_counts()
```

cough	count
Mild	62
Strong	38

dtype: int64

```
df['cough']=df['cough'].map({'Mild':0,'Strong':1})
```

```
df
```

	age	gender	fever	cough	city	has_covid
0	60	0	103.0	0	Kolkata	No
1	27	0	100.0	0	Delhi	Yes
2	42	0	101.0	0	Delhi	No
3	31	1	98.0	0	Kolkata	No
4	65	1	101.0	0	Mumbai	No
...	...	...	...	...	...	...
95	12	1	104.0	0	Bangalore	No
96	51	1	101.0	1	Kolkata	Yes
97	20	1	101.0	0	Bangalore	No
98	5	1	98.0	1	Mumbai	No
99	10	1	98.0	1	Kolkata	Yes

100 rows × 6 columns

Next steps: Generate code with df | New interactive sheet

```
df['has_covid'].value_counts()
```

has_covid	count
No	55
Yes	45

dtype: int64

```
df['has_covid']=df['has_covid'].map({'No':0,'Yes':1})
```

```
df.head()
```

	age	gender	fever	cough	city	has_covid
0	60	Male	103.0	Mild	Kolkata	0
1	27	Male	100.0	Mild	Delhi	1
2	42	Male	101.0	Mild	Delhi	0
3	31	Female	98.0	Mild	Kolkata	0
4	65	Female	101.0	Mild	Mumbai	0

Next steps: Generate code with df | New interactive sheet

```
df.filter(items=['gender','city'])
```

	gender	city
0	0	Kolkata
1	0	Delhi
2	0	Delhi
3	1	Kolkata
4	1	Mumbai
...	...	...
95	1	Bangalore
96	1	Kolkata
97	1	Bangalore
98	1	Mumbai
99	1	Kolkata

100 rows × 2 columns

```
df[['fever','has_covid']]
```

	fever	has_covid
0	103.0	NaN
1	100.0	NaN
2	101.0	NaN
3	98.0	NaN
4	101.0	NaN
...	...	...
95	104.0	NaN
96	101.0	NaN
97	101.0	NaN
98	98.0	NaN
99	98.0	NaN

100 rows × 2 columns

```
df.head(3)
```

	age	gender	fever	cough	city	has_covid
0	60	Male	103.0	Mild	Kolkata	0
1	27	Male	100.0	Mild	Delhi	1
2	42	Male	101.0	Mild	Delhi	0

Next steps: Generate code with df | New interactive sheet

```
df['fever']=df['fever'].astype(int)
```

```
df.head(3)
```

	age	gender	fever	cough	city	has_covid
0	60	0	103	0	Kolkata	NaN
1	27	0	100	0	Delhi	NaN
2	42	0	101	0	Delhi	NaN

Next steps: Generate code with df | New interactive sheet

```
df['age']=df['age'].astype(float)
```

```
df.head(3)
```

	age	gender	fever	cough	city	has_covid
0	60	Male	103.0	Mild	Kolkata	0
1	27	Male	100.0	Mild	Delhi	1
2	42	Male	101.0	Mild	Delhi	0

Next steps: Generate code with df | New interactive sheet

```
df['fever'][1:3]=50
```

```
/tmp/ipykernel_5620/736839710.py:1: FutureWarning: ChainedAssignmentError: be
You are setting values through chained assignment. Currently this works in ce
A typical example is when you are setting values in a column of a DataFrame,

df["col"][row_indexer] = value

Use `df.loc[row_indexer, "col"] = values` instead, to perform the assignment

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/s

  df['fever'][1:3]=50
/tmp/ipykernel_5620/736839710.py:1: SettingWithCopyWarning:
```

```
A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/s
  df['fever'][1:3]=50
```

```
df.head(3)
```

	age	gender	fever	cough	city	has_covid
0	60.0	0	103	0	Kolkata	NaN
1	27.0	0	50	0	Delhi	NaN
2	42.0	0	50	0	Delhi	NaN

Next steps: Generate code with df | New interactive sheet

```
df['city'][0]='jaipur'
```

```
/tmp/ipykernel_5620/935263761.py:1: FutureWarning: ChainedAssignmentError: be
You are setting values through chained assignment. Currently this works in ce
A typical example is when you are setting values in a column of a DataFrame,

df["col"][row_indexer] = value

Use `df.loc[row_indexer, "col"] = values` instead, to perform the assignment

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/s

  df['city'][0]='jaipur'
/tmp/ipykernel_5620/935263761.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/s
  df['city'][0]='jaipur'
```

```
df.head(3)
```

	age	gender	fever	cough	city	has_covid
0	60	Male	103.0	Mild	jaipur	0
1	27	Male	100.0	Mild	Delhi	1
2	42	Male	101.0	Mild	Delhi	0

Next steps: Generate code with df | New interactive sheet

```
df=df.rename(columns={'cough':'updated_cough'})
```

```
df=df.rename(columns={'age':'ages'})
```

```
df.head(3)
```

	age	gender	fever	cough	city	has_covid
0	60	Male	103.0	Mild	jaipur	0
1	27	Male	100.0	Mild	Delhi	1
2	42	Male	101.0	Mild	Delhi	0

Next steps: Generate code with df | New interactive sheet

```
df['updates_city']=df['city'].where(df['city']=='Delhi',other=0)
```

```
df.head(10)
```

	age	gender	fever	cough	city	has_covid	updates_city
0	60	Male	103.0	Mild	jaipur	0	0
1	27	Male	100.0	Mild	Delhi	1	Delhi
2	42	Male	101.0	Mild	Delhi	0	Delhi
3	31	Female	98.0	Mild	Kolkata	0	0
4	65	Female	101.0	Mild	Mumbai	0	0
5	84	Female	NaN	Mild	Bangalore	1	0
6	14	Male	101.0	Strong	Bangalore	0	0
7	20	Female	NaN	Strong	Mumbai	1	0
8	19	Female	100.0	Strong	Bangalore	0	0
9	64	Female	101.0	Mild	Delhi	0	Delhi

Next steps: Generate code with df | New interactive sheet

```
df['has_covid']=df['has_covid'].where(df['has_covid']==0,other='positive')
```

```
df.head(10)
```

	age	gender	fever	cough	city	has_covid	updates_city
0	60	Male	103.0	Mild	jaipur	0	0
1	27	Male	100.0	Mild	Delhi	positive	Delhi